Progress on Nash-Existence in Edge-Ranking Flow Allocation Games: A Rigorous Conditional Σ_2^p-Hardness Reduction

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Abstract

This note audits and repairs as much as possible of a proposed proof that pure-Nash existence in edge-ranking flow allocation games is Σ_2^p-complete. The conclusion is deliberately cautious. I do *not* obtain a complete solution of the original all-strategic problem. I do obtain a clean Σ_2^p-completeness theorem for a closely related fixed-priority variant in which some auxiliary vertices have prescribed rankings and are not strategic. The reduction fixes the two main logical problems in the submitted draft: inconsistent universal "assignments" are excluded by guard collectors, and the no-equilibrium alarm is used with a correct zero-signal equilibrium.

The remaining gap is exactly the one that the submitted draft tried, but failed, to close: converting the fixed-priority auxiliary vertices into ordinary strategic edge-ranking vertices without changing the clearing state or introducing new equilibria. The note includes explicit counterexamples to the submitted priority-locking lemma and explains why the conditional result below is partial progress rather than a solution of the authors' open problem.

1 Status of the result

The open problem asks for the exact complexity of deciding whether an edge-ranking flow allocation game admits a pure Nash equilibrium. The source paper proves polynomial-time computability of the clearing state for a fixed edge-ranking profile, proves strong NP-hardness of pure-Nash existence, and remarks that membership in Σ_2^p is easy while Σ_2^p-hardness remains open.

Outcome of this note.

(a) The original all-strategic problem ER–NE is verified to lie in Σ_2^p.

(b) The submitted proof of Σ_2^p-hardness is not correct. In particular, its priority-locking lemma and split tester are false as stated.

(c) A corrected reduction proves Σ_2^p-completeness for a fixed-priority extension FP–ER–NE of the problem. This is genuine partial progress: it gives a complete quantified-Boolean reduction once fixed auxiliary priorities are allowed.

(d) I do not claim that this proves Σ_2^p-hardness of ER–NE. The missing step is a valid fixed-priority-to-strategic implementation gadget.

2 Model and the basic upper bound

A flow allocation game consists of a directed graph $G = (V, E)$, capacities $c_e \geq 0$, and fixed supplies $b_v \geq 0$. In an edge-ranking game, each node v chooses a strict total order of its outgoing edge set $E^+(v)$. For a ranking profile π, a feasible flow f is a fixed point of the greedy allocation map: if node v has available supply

$$b_v + \sum_{e \in E^-(v)} f_e,$$

then v sends this supply along its outgoing edges greedily in the order π_v, subject to capacities. The selected clearing state $\hat{f}(\pi)$ is the coordinate-wise maximum feasible fixed point. Player v's utility is

$$u_v(\pi) = \sum_{e \in E^+(v)} \hat{f}_e(\pi).$$

A pure Nash equilibrium is a profile in which no single vertex can strictly increase its utility by changing its own outgoing-edge order.

Definition 1 (ER–NE)**.** *The input is an ordinary edge-ranking flow allocation game. The question is whether it has a pure Nash equilibrium.*

Proposition 1. ER–NE $\in \Sigma_2^p$.

Proof. A ranking profile has polynomial encoding length. Given a profile π, a player v, and an alternative ranking π'_v, the clearing states of π and (π'_v, π_{-v}) can be computed in polynomial time by the TopCycleIncrease algorithm of Bertschinger, Hoefer, and Schmand. Thus the predicate

$$u_v(\pi'_v, \pi_{-v}) \leq u_v(\pi)$$

is decidable in deterministic polynomial time. Therefore the statement

$$\exists \pi \ \forall (v, \pi'_v) \quad u_v(\pi'_v, \pi_{-v}) \leq u_v(\pi)$$

decides ER–NE and has the form of a Σ_2^p computation. □

3 Why the submitted all-strategic proof is not fixed by small edits

This section records two concrete failures of the submitted proof. They are included because they identify exactly what a complete solution still has to overcome.

3.1 The proposed priority lock changes logical flow

The submitted draft proposes to enforce a prescribed auxiliary order by adding large private return cycles. This does not preserve the original clearing state.

Proposition 2. *The priority-locking lemma in the submitted draft is false as stated.*

Proof. Consider a logical construction H with one auxiliary node w, no fixed supply, no incoming edges, and two logical outgoing edges e_1, e_2 of capacity 1 to sinks. In H, both logical edges carry zero flow in every clearing state.

Apply the submitted lock construction to enforce $e_1 \prec e_2$. It adds lock arcs λ_1, λ_2 with private return chains back to w, capacities L_1, L_2, and additional fixed supply $L = L_1 + L_2$ at w. In the intended order

$$\lambda_1, e_1, \lambda_2, e_2,$$

the private lock cycles can saturate λ_1 and λ_2 in the maximum clearing state. Their return flow comes back to w. Hence w has at least

$$L + L_1 + L_2 = 2L$$

available supply in the clearing state. After saturating the two lock arcs, w still has enough supply to send positive flow, indeed one unit each, on e_1 and e_2 whenever $L \geq 2$. Thus the logical flow is changed from zero to positive. This contradicts the preservation property required by the locking lemma. □

3.2 The proposed split tester allows inconsistent universal choices

The submitted split tester assumes that a tester ranking behaves like a legal universal assignment. Edge rankings do not enforce this.

Proposition 3. *The submitted split-tester lemma is false as stated.*

Counterexample. Use one universal variable y, two existential variables x, z, and set the existential assignment to $x = z = 1$. Let

$$F = (x \vee \neg z \vee y) \wedge (\neg z \vee \neg z \vee \neg z).$$

For every value of y, the second clause is false, so $F(1, 1, y)$ is unsatisfiable. In the submitted tester, however, the tester can spend flow on an unselected $z = 0$ track before spending flow on a true $x = 1$ literal. This primes the later $\neg z$ occurrence and then uses it to satisfy the second clause. Thus the tester can make both clause collectors fire even though no consistent universal assignment satisfies the residual formula. The problem is not the particular formula; it is the absence of a gadget forcing one legal value per universal variable. □

4 A corrected alarm lemma

The submitted draft also used the no-equilibrium module from Bertschinger, Hoefer, and Schmand incorrectly in the zero-signal case. With zero external supply, the maximum clearing state is not the zero flow: optional cycles can carry flow. Fortunately, the module still has the conditional behavior needed for a guarded reduction.

Let A be the seven-node directed graph with vertices $a_1, \ldots, a_7$ and arcs

arc	capacity
$a_5 \to a_2$	1
$a_2 \to a_5$	6
$a_2 \to a_1$	6
$a_4 \to a_2$	2
$a_1 \to a_4$	4
$a_1 \to a_6$	4
$a_6 \to a_3$	2
$a_3 \to a_1$	6
$a_7 \to a_3$	1
$a_3 \to a_7$	6.

Only a_1, a_2, a_3 have nontrivial choices. Node a_1 chooses between a_1a_4 first and a_1a_6 first; node a_2 chooses between a_2a_1 first and a_2a_5 first; node a_3 chooses between a_3a_1 first and a_3a_7 first.

Lemma 1 (conditional alarm)**.** *The alarm graph A has the following two properties.*

(i) If no external supply enters A, then A has a pure Nash equilibrium.

(ii) If two external units enter a_2 and two external units enter a_3, then A has no pure Nash equilibrium.

Proof. For (i), consider the profile

$$a_1a_4 \prec a_1a_6, \qquad a_2a_1 \prec a_2a_5, \qquad a_3a_7 \prec a_3a_1.$$

The maximum clearing state sends two units around the cycle

$$a_1 \to a_4 \to a_2 \to a_1$$

and one unit around the cycle

$$a_3 \to a_7 \to a_3.$$

Thus $(u_{a_1}, u_{a_2}, u_{a_3}) = (2, 2, 1)$. If a_1 switches its order, its utility becomes 0; if a_2 switches, its utility becomes 1; if a_3 switches, its utility becomes 0. Hence this profile is a PNE.

For (ii), direct computation of the maximum clearing state for the eight possible profiles gives the following payoff tables for (a_2, a_3) after fixing a_1's order. When $a_1a_4 \prec a_1a_6$,

	$a_3a_1 \prec a_3a_7$	$a_3a_7 \prec a_3a_1$
$a_2a_1 \prec a_2a_5$	$(4, 4)$	$(4, 3)$
$a_2a_5 \prec a_2a_1$	$(5, 2)$	$(3, 3)$.

Each cell has a profitable deviation by either a_2 or a_3. When $a_1a_6 \prec a_1a_4$, the symmetric table is

	$a_3a_1 \prec a_3a_7$	$a_3a_7 \prec a_3a_1$
$a_2a_1 \prec a_2a_5$	$(4, 4)$	$(2, 5)$
$a_2a_5 \prec a_2a_1$	$(3, 4)$	$(3, 3)$.

Again every cell has a profitable deviation. Hence no profile is a pure Nash equilibrium. □

5 A fixed-priority variant

The reduction below is for a variant in which some auxiliary nodes have prescribed rankings and are not strategic. This is not the original problem, but it captures the reduction skeleton that the submitted proof was trying to implement.

Definition 2 (FP–ER–NE)**.** *An instance consists of an edge-ranking flow allocation game together with a partition of vertices into strategic vertices and fixed-priority vertices. A fixed-priority vertex has a prescribed outgoing-edge order and is not allowed to deviate. A pure Nash equilibrium is defined with respect to unilateral deviations of strategic vertices only. The decision problem asks whether such an equilibrium exists.*

Proposition 4. FP–ER–NE $\in \Sigma_2^p$.

Proof. The proof is identical to Proposition 1, except that the universal quantifier ranges only over deviations of strategic vertices. □

6 The guarded QBF reduction for the fixed-priority variant

We reduce from

$$\textsc{Exists–Forall–Unsat-3CNF}: \qquad \exists X\ \forall Y\ \neg F(X,Y),$$

where $F = C_1 \wedge \cdots \wedge C_m$ is a 3-CNF formula over disjoint variable sets

$$X = \{x_1, \ldots, x_n\}, \qquad Y = \{y_1, \ldots, y_r\}.$$

This problem is Σ_2^p-complete. By harmless padding, assume $m, r \geq 1$.

Set

$$R = 5, \qquad K = m + r, \qquad B = R(m+1) + 1, \qquad S = rB.$$

The number R is the size of one logical block. A full threshold consists of K blocks: one block for each clause and one guard block for each universal variable. The extra unit in B is a dummy prefix that absorbs the one-unit return bonus and prevents it from helping to activate a further value splitter.

6.1 Variable selectors and splitters

For every existential variable x_i, create a strategic selector s_i with fixed supply B and two outgoing edges

$$s_i \to X_{i,0}, \qquad s_i \to X_{i,1},$$

each of capacity B. Ranking $s_i \to X_{i,b}$ first means choosing $x_i = b$.

Each existential splitter $X_{i,b}$ is fixed-priority. For every clause C_j satisfied by the literal $x_i = b$, add an edge

$$X_{i,b} \to c_j$$

of capacity R. Add one dummy edge from $X_{i,b}$ to a sink of capacity

$$B - R \cdot \#\{j : C_j \text{ is satisfied by } x_i = b\}.$$

The fixed order at $X_{i,b}$ puts the dummy edge first, followed by all clause edges in an arbitrary order. If $X_{i,b}$ receives its full block B, all its outgoing edges are saturated, so every satisfied clause receives a block of size R. If it receives at most one extra return unit, the dummy edge absorbs that unit.

There is one strategic tester vertex d with fixed supply $S = rB$. For each universal variable y_ℓ and value $b \in \{0,1\}$, add an edge

$$d \to Y_{\ell,b}$$

of capacity B. Add also a private bonus edge $d \to p$ of capacity B to a sink p.

Each universal splitter $Y_{\ell,b}$ is fixed-priority. It has one guard edge

$$Y_{\ell,b} \to q_\ell$$

of capacity R, and for every clause C_j satisfied by $y_\ell = b$, an edge

$$Y_{\ell,b} \to c_j$$

of capacity R. Its dummy edge has capacity

$$B - R\left(1 + \#\{j : C_j \text{ is satisfied by } y_\ell = b\}\right).$$

The fixed order puts the dummy edge first, followed by the guard and clause edges.

6.2 Collectors, threshold, and alarm

Each clause collector c_j is fixed-priority and has a single outgoing edge

$$c_j \to g$$

of capacity R. Each universal guard collector q_ℓ is fixed-priority and has a single outgoing edge

$$q_\ell \to g$$

of capacity R.

The threshold vertex g is fixed-priority. Its outgoing edges, in order, are

$$g \to h \quad \text{of capacity } R(K-1),$$

then

$$g \to a_2 \quad \text{of capacity } 2, \qquad g \to a_3 \quad \text{of capacity } 2, \qquad g \to d \quad \text{of capacity } 1.$$

The node h is a sink. Finally, attach the alarm graph A of Lemma 1; the alarm vertices a_1, a_2, a_3 are strategic, and the remaining alarm vertices have only one outgoing edge.

The strategically relevant vertices are therefore the existential selectors s_i, the tester d, and the alarm vertices a_1, a_2, a_3. All other nontrivial rankings are fixed.

Lemma 2 (block semantics)**.** *Fix rankings of the selectors and let α be the induced assignment of X. Let the tester d rank first exactly r edges of capacity B, none of which is the bonus edge, and let $T \subseteq \{(\ell, b) : \ell \in [r], b \in \{0,1\}\}$ be the corresponding selected universal literals. Then the following hold.*

(i) Guard q_ℓ sends a block of size R to g iff T contains at least one of $(\ell, 0), (\ell, 1)$.

(ii) Clause collector c_j sends a block of size R to g iff clause C_j is satisfied either by α or by one of the selected universal literals in T.

(iii) All $K = m + r$ collectors send blocks to g iff T contains exactly one value for every universal variable and the corresponding assignment β satisfies $F(\alpha, \beta)$.

Proof. A selected existential or universal splitter receives exactly B units. Its total outgoing capacity is exactly B, so all outgoing edges are saturated, independently of the order after the dummy edge. An unselected splitter receives no flow, except possibly the one-unit return bonus if it is immediately after the first r full B-edges in d's order. That one unit is absorbed by the dummy edge because every splitter has dummy capacity at least 1.

Thus a guard q_ℓ receives a block exactly when at least one value of y_ℓ is selected. A clause collector receives a block exactly when at least one selected literal, existential or universal, satisfies that clause. If all r guards receive blocks and d selected only r universal value edges, then exactly one value was selected for every universal variable. The m clause blocks then occur exactly when the resulting assignment satisfies all clauses. □

Lemma 3 (completeness)**.** *If there is an assignment $\alpha : X \to \{0,1\}$ such that $F(\alpha, Y)$ is unsatisfiable for every assignment of Y, then the constructed fixed-priority game has a pure Nash equilibrium.*

Proof. Choose each selector s_i to rank $s_i \to X_{i,\alpha_i}$ first. Let the tester d rank any one value edge for each universal variable first, and then rank the bonus edge $d \to p$. Put the alarm in the zero-signal equilibrium from Lemma 1(i).

We verify unilateral deviations. A selector always has exactly B fixed supply and two outgoing edges of capacity B, so its utility is B no matter which value it ranks first. The alarm receives no signal in the proposed profile, so no alarm vertex has a profitable deviation by Lemma 1(i).

Consider any deviation of the tester d. Since all outgoing edges of d have capacity B and d has fixed supply rB, the fixed supply fills exactly the first r edges in d's ranking. If one of these is the bonus edge, then at least one universal guard is missing and the threshold cannot be full. Otherwise, the first r edges select r universal literals. If all guards and all clauses fired, Lemma 2 would produce a genuine assignment β with $F(\alpha, \beta) = 1$, contrary to the assumption. Hence at most $K - 1$ collectors send blocks to g.

The first threshold edge $g \to h$ has capacity $R(K - 1)$. It absorbs every non-full collection of blocks. Therefore no tester deviation creates the signal edges $g \to a_2$, $g \to a_3$, or $g \to d$. The tester's utility is always at most its fixed supply $S = rB$, which it already attains. Thus no strategic vertex has a profitable deviation. □

Lemma 4 (soundness)**.** *If the constructed fixed-priority game has a pure Nash equilibrium, then the selector assignment α satisfies*

$$\forall Y\ \neg F(\alpha, Y).$$

Proof. Let π be a pure Nash equilibrium and read α from the first-ranked edge of each selector. Suppose, for contradiction, that there is an assignment β of the universal variables with $F(\alpha, \beta) = 1$.

Let d deviate by ranking the value edges $d \to Y_{\ell,\beta_\ell}$ first for all ℓ, and ranking the bonus edge $d \to p$ immediately after them. The r selected universal splitters each receive B units and therefore saturate their guard and all relevant clause edges. The selected existential splitters do the same. Since $F(\alpha, \beta) = 1$, all m clause collectors and all r guard collectors send blocks of size R to g. Thus g receives RK units.

The threshold edge $g \to h$ absorbs $R(K - 1)$ units. The remaining $R = 5$ units saturate the edges $g \to a_2$, $g \to a_3$, and $g \to d$, sending two units to each of a_2, a_3 and one unit back to d. Under the deviating ranking, d sends its fixed supply S on the selected value edges and sends the returned unit on its bonus edge. Hence d obtains utility $S + 1$.

In the original profile π, the same full threshold cannot already occur. If it did, the alarm would receive two units at a_2 and two units at a_3, and Lemma 1(ii) would imply that one of the alarm players has a profitable deviation, contradicting that π is a PNE. Therefore g does not send a return unit to d in π, and d's utility is at most its fixed supply S. The deviation above is strictly profitable, a contradiction. □

Theorem 1. FP–ER–NE *is Σ_2^p-complete.*

Proof. Membership is Proposition 4. The reduction from $\exists X \forall Y \neg F(X, Y)$ is polynomial in $|F|$. By Lemmas 3 and 4, the constructed fixed-priority game has a pure Nash equilibrium exactly when the quantified formula is true. Hence FP–ER–NE is Σ_2^p-hard and therefore Σ_2^p-complete. □

7 Why this remains partial progress for the original problem

Theorem 1 is not a proof of Σ_2^p-hardness for ER–NE, because ER–NE has no fixed-priority vertices. The fixed-priority vertices in the reduction are doing real work:

- the splitters use dummy-first rankings to make a full B-block all-or-nothing while absorbing the one-unit bonus;
- the threshold vertex g uses a prescribed order to suppress the signal unless all K logical collectors fire;
- the proof of completeness depends on these prescribed orders for *all* deviations of the tester.

The submitted draft attempted to remove fixed priorities using private return-cycle locks. Proposition 2 shows that this implementation changes the logical clearing state. A different implementation is needed. Merely observing that auxiliary splitters are often indifferent is also insufficient: an indifferent noncanonical splitter can route partial return flow into logical collectors, and the returned flow can then be recycled by the tester. This can create near-threshold profiles that are not represented by any Boolean assignment.

Thus the present note should be read as a rigorous reduction blueprint plus a precise isolation of the missing gadget. A complete solution of the original open problem would follow from a polynomial-size implementation of the fixed-priority vertices in Theorem 1 by ordinary strategic edge-ranking vertices, with two preservation properties: canonical fixed-priority behavior must be a best response, and noncanonical behavior must not hide a satisfying universal assignment or create a spurious equilibrium.

8 Summary

The original manuscript does address the authors' actual open problem, but its proof is not correct. The clean result obtained here is weaker:

$$\boxed{\text{FP–ER–NE is } \Sigma_2^p\text{-complete.}}$$

For the original all-strategic problem ER–NE, the verified status remains

$$\boxed{\text{ER–NE} \in \Sigma_2^p, \quad \text{ER–NE is strongly NP-hard, and } \Sigma_2^p\text{-hardness is not proved here.}}$$

The main mathematical progress is a corrected guarded QBF tester and a corrected use of the no-equilibrium alarm. The main unresolved obstacle is a sound priority-enforcement gadget for clearing-state edge-ranking games.

References

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